

Final Report & Literature Review

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1 Introduction

Semantic segmentation is a computer vision tool with the task of assigning a semantic label to each pixel of an image, or in other terms, defining boundaries between different semantic classes.

Semantic segmentation has been successfully applied in numerous different contexts, from medical imaging to self-driving cars [1].

In 2020 NASA has published an open dataset of images with segmentation masks from multiple Mars missions, taken by rovers currently operating on Mars.

Numerous groups have evaluated the capacity of different neural networks in executing semantic segmentation tasks on the AI4Mars dataset, including access time to prediction in the limited hardware of the rovers.

In this work we focus our effort into evaluating the choice of loss functions in image segmentation tasks in the m2020 navigational dataset from the AI4Mars dataset, containing NUMBER images and masks.

We also explore the effects of pre-training the model of choice with the MSL Navigational dataset, containing almost 30,000 images.

The model DeepLabV3+ with ResNet-50 was trained with three different loss functions: Cross Entropy, Dice Loss and the logarithmic cosine of the Dice loss.

In 3 training loops, the training started with weights resulting from a pretraining process in the MSL dataset, and in other 3 training loops, the model was just trained in the m2020 navigational dataset, starting from the model preset weights.

In part 2 of the work, we explore the literature on Image Segmentation and previous efforts regarding semantic segmentation on the AI4Mars dataset to motivate our work in the area.

In part 3 we describe in detail the training process and hyperparameter choices, and in part 4 we explore the results, specially in relation to different choices of loss function and pretraining.

2 Literature Review

2.1 Semantic Segmentation

Models

Loss Functions

2.2 AI4Mars

3 Training Method

3.1 Dataset and Preprocessing

The m2020 navigational dataset, consisting of `NUMBER` images, is split randomly into 90% for the training set and 10% for the validation set. Both datasets are resized, grayscaled, and converted into normalized tensors using the mean and variation suggested by the PyTorch implementation of the DeepLabV3+ model.

The training dataset additionally passes through an augmentation loop which includes:

- Horizontal flip (50% probability)
- Rotation up to 20° (50% probability)
- Random cutouts (up to 8, 50% probability)
- Random brightness adjustment (20% probability)

3.2 DeepLabV3+ with ResNet-50

DeepLabV3+ belongs to a family of neural networks that exploits dilated (atrous) convolutions to increase the field of view without increasing the number of parameters. While [3] was successful in implementing DeepLabV3+ on an independent dataset of Martian images, this work implements DeepLabV3+ with a ResNet-50 backbone on the open AI4Mars dataset.

We utilized the PyTorch implementation of DeepLabV3+, which employs Atrous Spatial Pyramid Pooling (ASPP) with an encoder-decoder topology [2]. The training process uses different initial learning rates: 0.0005 for the classifier (ASPP onwards) and 0.00001 for the ResNet-50 backbone.

The optimizer is Stochastic Gradient Descent (SGD) with a decay rate of 0.0001 and momentum of 0.9. A polynomial scheduler is used to dynamically decrease the learning rate with a power of 0.9.

3.3 Training Procedure

The model is trained for a total of 150 epochs using a progressive resizing strategy:

- **Epochs 1–50:** 128×128 resolution
- **Epochs 51–100:** 256×256 resolution
- **Epochs 101–150:** 512×512 resolution

The model was trained six different times to evaluate three possible choices of loss function: Cross Entropy Loss, Dice Loss, and Log-Cosh Dice Loss [1]. For each loss function, the model was trained under two conditions:

1. Starting from default weights (`DeepLabV3_ResNet50_Weights`).
2. Starting from a version pretrained on the `NUMBERj` images of the MSL navigational dataset using Cross Entropy loss.

4 Results

Model	Precision				Recall			
	Soil	Bedrock	Sand	Bigrock	Soil	Bedrock	Sand	Bigrock
ResNet-101	0.933	0.857	0.664	0.558	0.746	0.570	0.975	0.283
ResNet-101 (ImageNet)	0.999	0.467	0.667	0.618	0.596	0.958	0.952	0.355
ResNet-101 (MSL)	0.916	0.917	0.754	0.174	0.838	0.536	0.957	0.64

Table 1: Performance metrics for models trained on the M2020 dataset.

References

- [1] Reza Azad, Moein Heidary, Kadir Yilmaz, Michael Hüttemann, Sanaz Karimijafarbigloo, Yuli Wu, Anke Schmeink, and Dorit Merhof. Loss functions in the era of semantic segmentation: A survey and outlook. (arXiv:2312.05391), December 2023. arXiv:2312.05391 [cs].
- [2] Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam. Rethinking atrous convolution for semantic image segmentation. (arXiv:1706.05587), December 2017. arXiv:1706.05587 [cs].
- [3] Cristina Luna, Robert Field, and Steven Kay. Ai-enabled capabilities to facilitate next-generation rover surface operations. (arXiv:2510.05985), October 2025. arXiv:2510.05985 [cs].